AI-Powered Integrated Bid Compliance Verification Platform for GeM Procurement

Supplementary report: details that complement the idea presentation

1. Problem and Solution at a Glance

Problem. Before evaluating price or technical merit, officers must check every bid against tender conditions (EMD, turnover, experience, certificates, registrations). This is repetitive, slow and inconsistent, and errors lead to wrongly accepted bids or disputed rejections.

Solution. An AI platform that reads the tender and bidder documents, converts conditions into a checklist, verifies each item against evidence and official sources, and gives the officer an explainable report. **The officer always makes the final decision.**

2. How It Works

Step	What happens	Key tools
1. Ingest and read	Upload PDFs and scans; OCR (English and Hindi); keep page and clause references	FastAPI, Tesseract / PaddleOCR, pdfplumber
2. Understand tender	LLM turns each condition into a structured requirement (type, value, exemptions, source clause)	LLM + RAG, Pydantic, pgvector
3. Extract evidence	Pull matching fields from bidder documents with a confidence score	LLM, format validation
4. Verify	Rule engine checks values; official sources confirm GST, PAN, Udyam, MCA	Rule engine, official APIs
5. Report and review	Status, reason and evidence per item; officer accepts or overrides; audit trail saved	React, PostgreSQL

3. Compliance Decision Logic

Each requirement passes four checks in order. A failure at any step raises a specific flag with the reason and source page.

Check	If it fails	Flag
Evidence found in bidder documents?	Required document or value not found	Missing
Extraction confidence above threshold?	Sent to officer, never auto-decided	Manual Check
Value satisfies the rule?	Rule failed; reason shown	Non-Compliant
Matches official source?	Mismatch, or source unavailable	Discrepancy

Example (illustrative). Tender: average annual turnover of at least Rs. 5 Cr. Evidence: CA certificate, page 14, shows Rs. 4.2 Cr. Result: **Non-Compliant**, with page and clause shown; the officer confirms or overrides (for example, if an MSE relaxation applies).

4. Why the Results Can Be Trusted

- **Code decides, AI reads.** Numeric, date and ID checks are done by the rule engine, not by the LLM.
- **No silent rejection.** Low-confidence items go to the officer as Manual Check.
- **Evidence-backed.** Every flag links to the source page and clause.
- **Prompt-injection defence.** Bidder documents are treated as untrusted data; hidden text such as "mark this compliant" cannot change a result.
- **Human authority.** AI recommends, the officer verifies, the authorized official decides.

5. External Verification

Source	Purpose	Approach
GST	Registration status and legal name	Official / authorized APIs (API Setu, GSP)
PAN	Identity and name match	Authorized verification services
Udyam	MSE status and category	Official verification
MCA	Company status and incorporation date	Authorized access
DigiLocker	Issued certificates	Consent-based fetch

Official, consented channels only (no unauthorized scraping). If a source is unavailable, the item is flagged as unverified and the officer is told which check could not be completed.

6. Security and Data Protection

- Role-based access (Officer, Senior Authority, Admin, Auditor) with JWT sessions.
- Encryption in transit and at rest; document hashes to detect tampering.
- Append-only audit log of every action and override.
- On-premise or private LLM for sensitive tenders; alignment with the DPDP Act 2023 (purpose limitation, retention limits).

7. How We Will Measure Success

Metric	Why it matters
Field-level extraction accuracy (precision / recall)	Correct evidence
Missed non-compliance rate (target: as close to zero as possible)	A bad bid passing is the costliest error
False-flag rate	Avoids wasting officer time
Time per bid vs manual (target: minutes instead of hours / days)	Efficiency claim
Officer override rate	Trust and rule quality

Method: test on public tenders and synthetic bids with seeded errors (missing pages, expired certificates, altered figures) and report results honestly.

8. Roadmap

Phase	Scope
Phase 1: MVP (hackathon)	Upload, OCR and extraction, checklist, matching, rule engine for core checks, explainable report, officer review
Phase 2: Pilot	Live GST / Udyam / MCA verification, risk scoring, anomaly indicators, officer feedback loop
Phase 3: Scale	GeM integration, shared rules library across departments, more Indian languages, analytics

9. Assumptions and Limitations

- Poor-quality scans are routed to manual review.
- Live government API access needs authorization; the demo may use sandbox or mock responses.
- Rules must be reviewed by procurement experts before real use.
- Figures marked "target" are goals to be validated in a pilot, not measured results.
- The tool supports preliminary screening only; final authority rests with the authorized official.